

Total No. of Questions : 6]

SEAT No. :

P44

[Total No. of Pages : 2

APR - 18/TE/Insem. - 46

T.E. (Information Technology)

DESIGN & ANALYSIS OF ALGORITHMS

(2012 Course) (Semester - II) (314449)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Write a general plan for analyzing time efficiency of Recursive Algorithm. [5]
b) Prove by method of contradiction that $2 + \sqrt{3}$ is an irrational number. [5]

OR

- Q2)** a) Solve the following recurrence relation using substitution method. [5]
 $T(n) = T(n-1) + n$, $n > 0$
 $T(0) = 0$, $n = 0$
b) Explain the Aggregate method of Amortized analysis with example. [5]

- Q3)** a) Write a recurrence relation for quick sort and find out its time complexity by substitution method. [5]
b) Solve the following instance of job sequencing problem using greedy approach. Let $n = 4$, profit $p(1:4) = (60, 30, 40, 80)$ and deadlines $d(1:4) = (2, 1, 2, 1)$. [5]

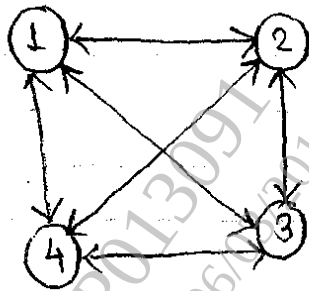
OR

- Q4)** a) Perform multiplication of given large integers 981×1234 in time less than $O(n^2)$. [5]
b) Solve the following instance of fraction knapsack problem using greedy approach. Let knapsack capacity $w = 20$ [5]

Items	Weight	Profit
1	14	24
2	18	20
3	10	16

P.T.O.

- Q5) a)** Find the solution of the following Travelling salesman problem using Dynamic approach. [6]

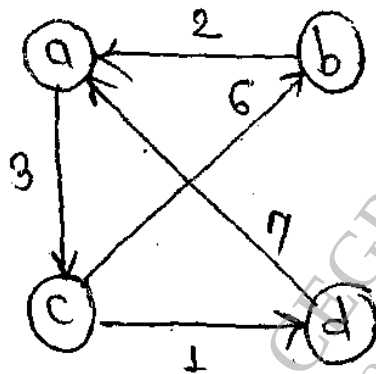


$$\begin{bmatrix} 0 & 10 & 15 & 20 \\ 5 & 0 & 9 & 10 \\ 6 & 13 & 0 & 12 \\ 8 & 8 & 9 & 0 \end{bmatrix}$$

- b) Compare dynamic programming with greedy approach. [4]

OR

- Q6) a)** Apply Floyd's algorithm to solve the all pairs shortest path problem for the given graph. [6]



- b) State "Principle of optimality". [4]

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